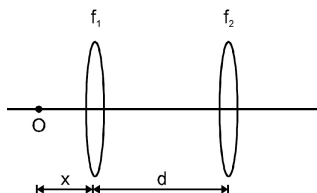
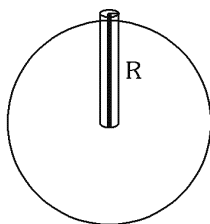


1. Two converging lenses have focal length f_1 and f_2 ($f_1 > f_2$). The optical axis of the two lenses coincide. This lens system is used to form an image of real object. It is observed that final magnification of the image does not depend on the distance x . Whole arrangement is shown in figure. Final magnification is:

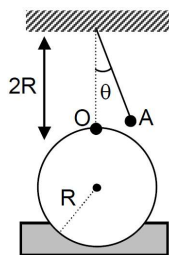


- (A) $-\frac{f_1}{f_2}$ (B) $-\frac{f_2}{f_1}$ (C) $-\frac{f_1}{f_1 + f_2}$ (D) $-\frac{f_2}{f_1 - f_2}$

2. Imagine a hole drilled along the radius of the earth. A uniform rod of length equal to the radius (R) of the earth is inserted into this hole. The distance of centre of gravity of the rod from the centre is

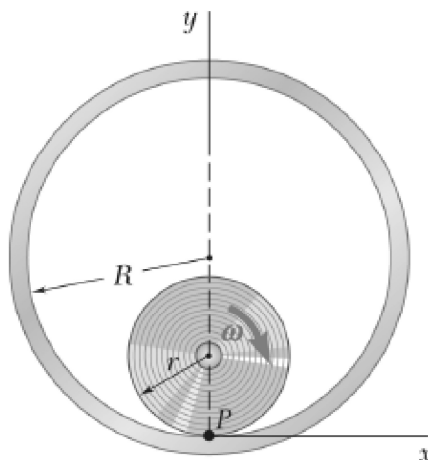


- (A) $\frac{R}{3}$ (B) $\frac{R}{2}$ (C) $\frac{2R}{3}$ (D) None of these
3. A uniformly charged non conducting sphere (in volume) of radius R and total charge Q is fixed on a nonconducting platform. A very small bob having negative charge $-q$ and mass m is hung from a fixed point through a non-conducting and inextensible string of length $2R$ as shown in the figure. The whole system is placed in a gravity free space. The bob is displaced by a very small angle from the vertical, then find time required to reach from point A to O. (Angular displacement of the bob is so small that it can be assumed, the distance between the bob and the sphere is always nearly equal to R)

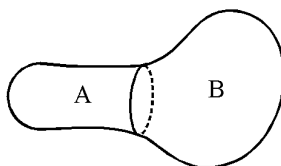


- (A) $2\pi\sqrt{\frac{4\pi\epsilon_0 R^3 m}{3Qq}}$ (B) $\frac{\pi}{2}\sqrt{\frac{4\pi\epsilon_0 R^3 m}{3Qq}}$ (C) $2\pi\sqrt{\frac{8\pi\epsilon_0 R^3 m}{3Qq}}$ (D) $\frac{\pi}{2}\sqrt{\frac{8\pi\epsilon_0 R^3 m}{3Qq}}$

4. A wheel of radius r rolls without slipping along the inside of a fixed cylinder of radius R with a constant angular velocity ω . Denoting by P the point of the wheel in contact with the cylinder at $t = 0$, horizontal components of the velocity of P at any time t . (The curve described by Point P is a hypocycloid.)

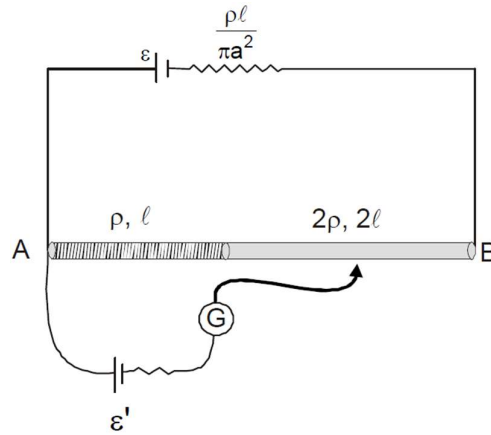


- (A) $\omega r \left(\cos \frac{\omega r t}{R-r} + \cos \omega t \right)$
- (B) $\omega r \left(\cos \frac{\omega r t}{R-r} - \sin \omega t \right)$
- (C) $\omega r \left(\cos \frac{R-r}{\omega r t} - \cos \omega t \right)$
- (D) $\omega r \left(\cos \frac{\omega r t}{R-r} - \cos \omega t \right)$
5. Figure shows an arbitrary body through which electric charge is distributed in an arbitrary way. The body is divided in two region A and B. Using the result.
- The electric force that the charge in region A exerts on the charge in region B is equal to the force that the charge on the entire body exerts on the charge in region B or otherwise solve next two problems.

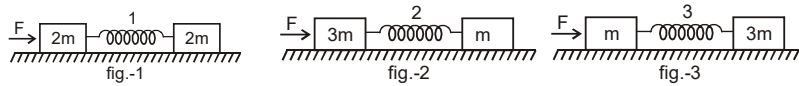


- (A) Consider a uniformly charged spherical shell of radius R having total charge Q . Electric force $\vec{F}$ exerted on each hemisphere by the charge on the other hemisphere is $\frac{Q^2}{32\pi\epsilon_0 R^2}$
- (B) Consider a uniformly charged spherical shell of radius R having total charge Q . Electric force $\vec{F}$ exerted on each hemisphere by the charge on the other hemisphere is $\frac{Q^2}{16\pi\epsilon_0 R^2}$
- (C) Consider a uniformly charged solid dielectric sphere of total charge Q and radius R . The force exerted by one hemisphere on other hemisphere is $\frac{3Q^2}{16\pi\epsilon_0 R^2}$
- (D) Consider a uniformly charged solid dielectric sphere of total charge Q and radius R . The force exerted by one hemisphere on other hemisphere is $\frac{Q^2}{9\pi\epsilon_0 R^2}$

6. In a potentiometer primary cell has internal resistance of $\frac{\rho\ell}{\pi a^2}$ as shown in figure. The potentiometer wire is combination of two wires connected in series as shown in figure. The common cross sectional radius of both wires used for potentiometer is 'a'.

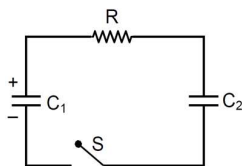


- (A) Range of potentiometer is $\frac{5\varepsilon}{6}$
- (B) Range of potentiometer is $\frac{2\varepsilon}{3}$
- (C) If null point is at mid point of longer wire then $\varepsilon' = \frac{\varepsilon}{2}$
- (D) If null point is at mid point of longer wire then $\varepsilon' = \frac{\varepsilon}{4}$
7. In each of three figures shown, two blocks are connected by a light spring and the system is placed on smooth horizontal surface. A constant horizontal force of magnitude F is applied to left block as shown. Assuming spring constant in all three cases to be same, which of the following statements is/are true.

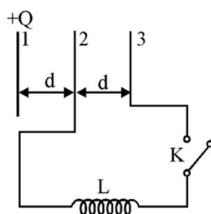


- (A) maximum compression in spring 1 is greater than that in spring 2.
- (B) maximum compression in spring 3 is greater than that in spring 1.
- (C) maximum compression in spring 3 is greater than that in spring 2.
- (D) maximum compression in all springs is equal.

8. In the circuit shown in figure $C_1 = C_2 = C$ and capacitor C_1 is having initial charge Q_0 . The switch is closed at $t = 0$. Then choose the correct option(s) :



- (A) The current at time t in the circuit will be $\left(\frac{Q_0}{RC}\right)e^{-2t/RC}$
- (B) Charge on capacitor C_1 at time t will be $\frac{Q_0}{2}(1 + e^{-2t/RC})$
- (C) Total heat loss in the circuit till the current becomes zero is $\frac{Q_0^2}{2C}$
- (D) Total heat loss in the circuit till the current becomes zero is $\frac{Q_0^2}{4C}$
9. A particle of mass m moving with constant speed u inside a fixed smooth spherical bowl of radius a describes a horizontal circle at a distance $\frac{a}{2}$ below its centre.
- (A) The radius of the circular motion is $\frac{a\sqrt{3}}{2}$
- (B) The value of u is $\sqrt{\frac{3ag}{2}}$
- (C) The normal reaction of the spherical surface on the particle is $\frac{mg}{2}$
- (D) The magnitude of the resultant force acting on the particle is zero, in an inertial frame.
10. Three identical fixed metal plates are arranged in the air at equal distances d from each other. The area of each of the plates equals A . The plate 1 has charge $+Q$, while plates 2 and 3 are not charged and are connected through a key K to the inductor L . The distance between the plates is small compared with their size. The ohmic resistance of the coil can be neglected :



- (A) Just before the switch K is closed, the charge on right surface of plate 2 is $\frac{Q}{2}$
- (B) Just before the switch K is closed, the charge on right surface of plate 3 is $\frac{Q}{2}$
- (C) After the switch K is closed, the maximum current in the circuit is $\frac{Q}{2}\sqrt{\frac{d}{L\epsilon_0 A}}$
- (D) After the switch K is closed, the maximum current in the circuit is $Q\sqrt{\frac{d}{L\epsilon_0 A}}$

11. In the shown figure (1) and (2), capacitors are in the steady state. Charging batteries are removed and switches S_1 and S_2 are closed at time $t = 0$. The plot of $\ell \ln I$ (I is the current in the resistor) in against time t in the resistors R_1 and R_2 are shown by the graphs 1 and 2 respectively in the figure (3). Choose the correct option(s).

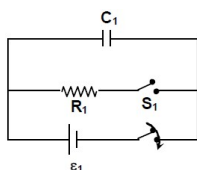


figure (1)

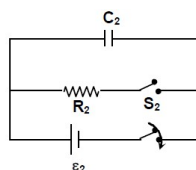


figure (2)

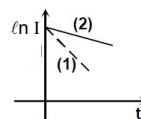
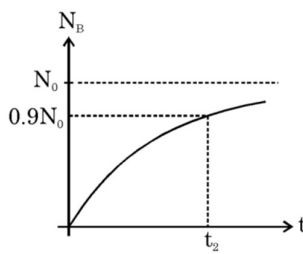
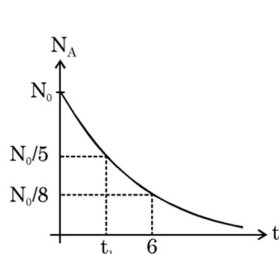


figure (3)

- (A) If $\varepsilon_1 = 2\varepsilon_2$, R_1 must be equal to $2R_2$ (B) If $R_1 = R_2$, C_1 must be less than C_2
 (C) If $C_1 = C_2$, R_1 must be less than R_2 (D) R_1C_1 is equal to R_2C_2
12. In a decay process A decays into B. Two graphs of number of nuclei of A and B versus time are shown. Mark the **CORRECT** statement(s) :

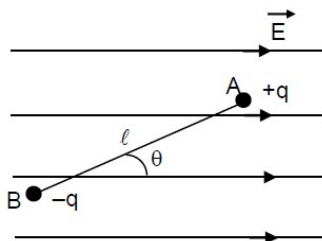


- (A) $t_2 - t_1 = 4$ (B) $t_2 - t_1 = 2$ (C) $t_1 = 2 \log_2 5$ (D) Halflife so of A is 2sec.

COMPREHENSION

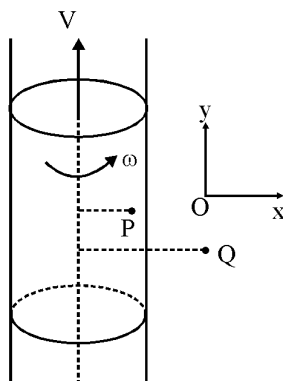
An ideal dipole of length ℓ and having charges A and B is released from the position shown in figure.

Given that $m_A = m$, $m_B = 2m$, $q_A = +q$, $q_B = -q$ and $\vec{E} = \text{uniform}$.



13. For $\theta = \frac{\pi}{2}$, which of the following options is/are correct :
- (A) Dipole will oscillate but not simple harmonically
 (B) Distance travelled by particle A upto the instant when its speed is maximum is $\frac{\pi\ell}{4}$
 (C) Initial angular acceleration of dipole is $\frac{3qE}{2m\ell}$

- (D) maximum angular speed of dipole during the motion is $\sqrt{\frac{3qE}{m\ell}}$
14. For $\theta = \theta_0$, (very small), Which of the following options is/are correct :
- (A) Maximum angular speed during the motion is independent of θ_0 .
- (B) Angular frequency of oscillations is independent of θ_0 .
- (C) Maximum speed of particle –A during the motion is $\theta_0 \sqrt{\frac{2qE\ell}{3m}}$
- (D) Maximum speed of particle –B during the motion is $\theta_0 \sqrt{\frac{qE\ell}{6m}}$
15. Consider a hollow pipe of infinite length and radius R. The pipe has uniform distribution of charge on its surface. The pipe is rotating with angular velocity ω about its axis. The pipe is also moving with velocity v along its length. There are two points P and Q as shown in the figure. Point P is inside the pipe and point Q is outside the pipe. Then match the component of magnetic field in Column–I with description given in Column–II.

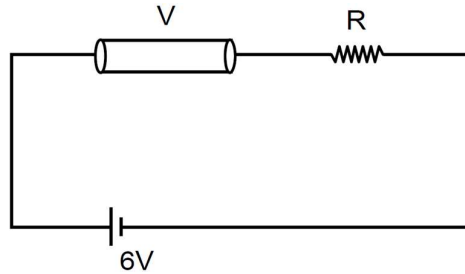


List-I		List-II	
(P)	B_x at point P	(1)	is zero
(Q)	B_x at point Q	(2)	depends on ω
(R)	B_y at point P	(3)	depends on v
(S)	B_z at point Q	(4)	inversely proportional to distance from the of pipe

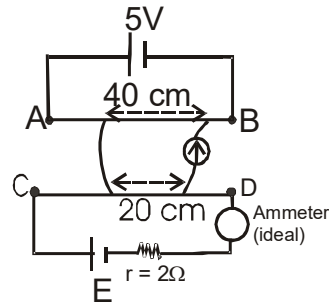
Codes:

	P	Q	R	S
(A)	1	1	2	3
(B)	1	2	1	3,4
(C)	1	2	3,4	3
(D)	1	1	2	3,4

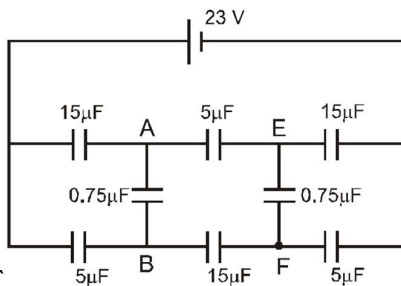
16. The Current developed in a non ohmic conductor is given $I = aV^{1/2}$, where V is the potential difference across the conductor. When it is connected with a resistance of $R\Omega$ in series with a battery of 6Volt as shown in the figure , the power developed in the conductor is twice the power generated is resistance of $R\Omega$. Find the value of R [$a = 0.2 \text{ A Volt}^{-1/2}$]



17. AB and CD are two uniform resistance wires of lengths 100 cm and 80 cm respectively. The connections are shown in the figure. The cell of emf 5 V is ideal while the other cell of emf E has internal resistance 2Ω . A length of 20 cm of wire CD is balanced by 40 cm of wire AB. Find the emf E in volt, if the reading of the ideal ammeter is 2 A. The other connecting wires have negligible resistance.

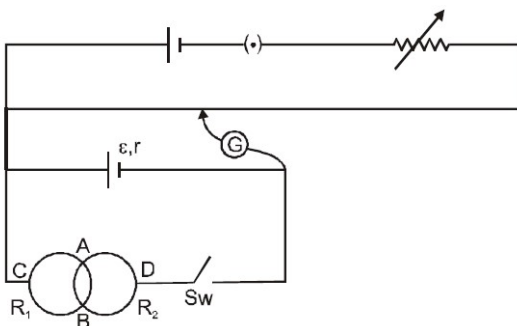


18. Find the potential difference (in volt) between the points A and B of the circuit shown in figure.



19. A particle of mass 2 kg is subjected to a two dimensional conservative force given by, $F_x = -2x + 2y$, $F_y = 2x - y^2$. (x, y in m and F in N). If the particle has kinetic energy of $(8/3)$ J at point (2,3), find the speed of the particle when it reaches (1, 2).

20. Internal resistance of a cell is $r = 5\Omega$ and it is connected in potentiometer (wire length = 1m) circuit arrangement as shown in figure, which shows two circular conducting rings R_1 & R_2 each having radii 20 cm crossing each other at conducting joints A & B. Section AB subtends 120° at the center of each ring. Resistance per unit length of R_1 & R_2 are $\frac{3}{\pi} \Omega/\text{cm}$ & $\frac{6}{\pi} \Omega/\text{cm}$ respectively. Cell is connected across points C & D of rings lying on perpendicular bisector of AB. Now when switch Sw remains closed, balancing length at null point is 48 cm. Find the balancing length (in cm.) when switch Sw is opened.



ANSWER KEY

1.	(B)	2.	(C)	3.	(B)	4.	(D)	5.	(A)
6.	(AC)	7.	(ABC)	8.	(ABD)	9.	(AB)	10.	(ABC)
11.	(ABC)	12.	(BCD)	13.	(ACD)	14.	(BCD)	15.	(D)
16.	5	17.	(12.00)	18.	5	19.	(02.00)	20.	52